MIL-C-10116C(EA)
31 January 1980
SUPERSEDING
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7 November 1969

MILITARY SPECIFICATION

CANISTER, M11

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of protective mask canister.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

STANDARDS

FEDERAL

FED-STD-101 - Preservation, Packaging and Packing Materials, Test Procedures.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-address Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

C5-41-11 - Packaging, Canister (For M11 Canister) Assembly.
D5-3-588 - Canister, M11.
C26-19-51 - Q113 Rough Handling Machine.
C38-5-296 - Q134 Container Sorbent Volumeter.
B38-6-351-8 - Q112 Canister Resistance Indicator.
B136-41-600-1 - Q95 All Purpose Gas Life Testing Apparatus.
B136-41-850-3 - Q216 Test Apparatus Agent Gas Sorption.
B136-42-102-3 - Q127 DOP Filter Testing Penetrometer.

3. REQUIREMENTS

3.1 Material. All materials and components used in the manufacture of the canister shall comply with the requirements of the respective specifications and drawings forming a part of this specification.

3.2 Manufacture. The canister shall be filled and assembled as follows: With the lower sorbent-retainer assembly in place fill the sorbent container with 250 ± 10 ml of impregnated charcoal. The surface of the charcoal in the container shall be essentially level at the end of the filling operation (see 4.4.4.1). Place the upper sorbent-retainer assembly in the container, with the filtering disc resting evenly on the top of the charcoal, and apply a compression load of 350 ± 25 pounds to the periphery of the upper sorbent-retainer to compress the charcoal. While maintaining the compression load on the charcoal, fasten the upper sorbent-retainer assembly in position by beading to the container wall. The beading shall be done in such a manner as to maintain the compression load on the charcoal. Assembly the filter element as shown on Drawing C5-3-583 and subsidiary drawings and complete the canister assembly as shown on Drawing D5-3-588. Weigh the canister, without closures, on a scale graduated in units of 1.0 gram, reading to the

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closest graduation. Mark the canister with the lot number and the previously determined weight as shown on Drawing D5-3-588. Close the canister to the atmosphere (see 6.4).

3.3 Loss in weight on heating (moisture content). The average loss in weight of the charcoal from completed canisters shall not exceed 4.0 percent by weight and the loss in weight of the charcoal from any individual canister shall not exceed 5.0 percent when tested as specified in 4.4.4.2.

3.4 Airflow resistance. The airflow resistance of the assembled canister, measured as a pressure differential, shall not exceed 75 mm of water and shall not be less than 50 mm of water when tested as specified in 4.4.4.3.

3.5 DOP smoke penetration. The canister shall be tested for DOP smoke penetration before and after rough handling (see 4.4.4.5) as specified in 4.4.4.4. DOP smoke penetration shall not exceed 0.025 percent before rough handling nor shall it exceed 0.050 percent after rough handling.

3.6 Gas life. The computed gas life mean minus 1.1 times the standard deviation shall be equal to or greater than the values shown in table I when testing the canister for gas life as specified in 4.4.4.6.

TABLE I. Gas life

Gas	Minimum life
GB 1/	180.0 min
CK 2/	30.0 min
DMMP 3/	150.0 min

1/ GB Gas Life - The break point shall be the time needed to attain an effluent concentration equal to 0.04 mg/m^3.

2/ CK Gas Life - The break point is defined as the time needed to attain an effluent concentration of 8.0 mg/m^3.

3/ DMMP Gas Life - The break point shall be the time needed to attain an effluent concentration equal to 0.04 mg/m^3.

3.7 Canister leakage. The assembled canister shall not leak in excess of 5 ml per minute when subjected to an air pressure of 5.5 pounds per square inch when tested as specified in 4.4.4.7.

3.8 Package leakage. The container body and top shall not leak when seamed as shown on Drawing C5-41-11 and tested as specified in 4.4.4.8.

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3.9 Workmanship. The canister shall be free from damage, defects, and contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Government's responsibility. The Government will be responsible for the performance of the GB test in 3.6. Samples shall be forwarded to the laboratory designated by the contracting officer (see 6.2).

4.1.3 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 500 canisters shall be produced using the same methods, materials, and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. Twenty-five canisters from the preproduction sample shall be examined for all requirements of the drawings. One hundred and twenty-five of the remaining canisters, taken at random, shall be examined in accordance with 4.4.3.4, except that the acceptance number shall be zero.

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4.3.2.2 For destructive tests. Representative canisters shall be taken from each preproduction lot and tested in accordance with table II. Twenty-five (25) representative canisters shall be taken from each preproduction lot and forwarded to a Government test agency, as specified in the contract, for GB life tests as specified in 4.4.4.6.

TABLE II. Preproduction testing

Test	Before rough handling	After rough handling
Volume of filling and charcoal moisture content	15	-
DOP penetration	All	25
CK gas life	25	-
DMMP life	13	12
GB gas life	25	-

4.3.2.3 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.3.2.1 and 4.3.2.2 to be acceptable. If the computed gas life mean minus 1.1 times the standard deviation fails to meet the requirements of 3.6, the lot represented shall be rejected. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the canisters produced by one manufacturer, at one plant, from the same materials under essentially the same manufacturing conditions, and during one days' production, except that no more than one lot of impregnated charcoal or one lot of filter elements may be represented in one lot of canisters.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. All unit pack canisters (5.1) shall be leak tested.

4.4.2.2 For destructive tests. Sampling shall be in accordance with table III.

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TABLE III. Regular production testing

Test	Before rough handling	After rough handling
Volume of filling charcoal moisture content	8	-
DOP penetration	All	16
CK gas life	16	-
DMMP	8	8

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample canisters shall be examined in accordance with the classification of defects and MIL-STD-105. All unit packed canisters (5.1) shall be leak tested as specified in 4.4.4.8.

4.4.3.2 For destructive tests. Sample canisters shall be tested as specified in 4.4.4.

4.4.3.3 Critical defects. Each canister in every lot shall be inspected for critical defects.

4.4.3.4 Classification of defects.

(a) Container, sorbent (Dwg. B5-3-843).

Categories		Defects
Critical:	None defined	
Major:	AQL 0.65 percent defective	
101		Component missing or incorrectly assembled
102		Retainer, burrs on element side
103		Beading incorrect
104		Depth to retainer (17/32) incorrect
105		Damage

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Minor: AQL 1.0 percent defective

201 Interior contamination (without disassembly)

AQL 2.5 percent defective

202 Exterior contamination

(b) Canister, M11 (Dwg. D5-3-588).

Categories	Defects	Acceptance standards
Critical:		
1	DOP smoke penetration (before rough handling)	4.4.4.4
2	Canister leakage	4.4.4.7
Major:	AQL 0.65 percent defective	
101	Airflow resistance	4.4.4.3
	AQL 1.0 percent defective	
102	Rolled thread major diameter incorrect	
103	Rolled thread minor diameter incorrect	
104	Canister damaged, preventing proper functioning	
105	Marking incorrect	
106	Weight incorrect	
107	Component assembled incorrectly	
108	Diameter at top of threaded neck	
109	Length of effective thread	
Minor:	AQL 2.5 percent defective	
201	Coating marred or scratched	
202	Marking missing or illegible	
203	Damged (not affecting functioning)	
204	Height of threaded neck	
	AQL 10 percent defective	
205	Contamination	

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4.4.3.4 Classification of defects (continued).

(c) Packaging inspection (see Section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 0.25 percent defective
101	Container not correct
102	Container not supplied with tear strip
103	Corrosion resistant key not furnished and/or not attached to end of each can lid as specified
	AQL 2.5 percent defective
104	Packing incorrect
105	Marking missing, incorrect or illegible

4.4.4 Tests.

4.4.4.1 Volume of filling. Volume of filling of the canister shall be determined with the Q134 Sorbent Container Volumeter (C38-5-296).

4.4.4.2 Loss in weight on heating (moisture content). The charcoal shall be removed from the sample canister and tested as follows:

(a) Weigh a 10 gram specimen of the charcoal, to the nearest milligram, in a tared flat form weighing bottle approximately 3 inches in diameter.

(b) Dry the specimen for 3 hours in an oven at a temperature of 150° ± 5° C. Cool in a desiccator and weigh.

(c) Calculate the loss in weight as follows:

$$\text{Percent loss} = \frac{100\ A}{W}$$

where: A = Loss in weight on heating
W = Original weight of specimen.

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4.4.4.3 Airflow resistance. Airflow resistance of the canister shall be determined, at an airflow rate of 85 liters per minute, with the Q112 Canister Resistance Indicator (B38-6-351-8).

4.4.4.4 DOP smoke penetration. DOP smoke penetration of the canister shall be determined, at a flow rate of 32 liters per minute, with the Q127 DOP Filter Testing Penetrometer (B136-42-102-3).

4.4.4.5 Rough handling. Rough handling of the canister shall be performed with the Q113 Rough Handling Machine (C26-19-51). Four canisters shall be rough handled on the machine at one time for 10 min + 5 seconds. Canisters to be given DOP penetration test after rough handling shall be packaged in the Packaging, Canister (C5-41-11) prior to being placed in the rough handling machine. These canisters shall not be painted before packaging. Canisters to be given gas life test after rough handling shall not be packaged.

4.4.4.6 Gas life. CK gas life of the canisters shall be determined with the Q95 All Purpose Gas Life Testing Apparatus (B136-41-600-1). The DMMP and GB gas life of the canister shall be determined with the Q216 Test Apparatus Agent Gas Sorption (B136-41-850-3). The canister shall be mounted so that axis of flow is in a horizontal direction. The conditions for gas life testing shall be as follows:

Gas	Concentration	Temperature	RH	Rate of airflow
GB	4.0 ± 0.4 mg/liter	26 ± 6°C	15%	32 liters/min[1]
CK	4.0 ± 0.5 mg/liter	26 ± 6°C	80%	50 liters/min[2]
DMMP	3.0 ± 0.2 mg/liter	26 ± 6°C	15%	50 liters/min[1]

1 Constant flow
2 Intermittent flow.

4.4.4.7 Canister leakage. The canister shall be tested using a suitable device which subjects the canister to the required air pressure (see 3.7) and evaluates the leakage flow rate.

4.4.4.8 Unit packing container leakage. The leakage characteristics of the sealed container shall be determined by subjecting the samples taken in accordance with 4.4.2.1 to the leakage test by either the hot water or vacuum chamber technique of FED-STD-101, method 5009.

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4.4.5 Acceptance/rejection criteria. If any sample canister fails volume of filling, loss in weight on heating, DOP penetration before rough handling, or the canister leakage before rough handling test, the lot represented shall be rejected. If more than two of the sixteen canisters subjected to DOP penetration after rough handling fail that test, the lot represented shall be rejected. If the computed gas life mean time 1.1 times the standard deviation fails to meet the requirement of 3.6 the lot represented shall be rejected.

5. PACKAGING

5.1 Preservation, level A. The canister shall be packaged as shown on Drawing C5-41-11.

5.1.1 Leakage test. The filter canister, unit packed as specified in 5.1, shall show no sign of leakage as evidenced by release of a continuous stream of bubbles which appear at any surface when tested as specified in 4.4.4.8. The container that fails the leakage test shall be repacked and retested.

5.2 Packing, level A, B or commercial (see 6.2).

5.2.1 Level A. Eighteen canisters, packaged as specified in 5.1, shall be packed in two layers (3 by 3 pattern) within a class 2, style 2, 2-1/2, or 4 nailed wood box, conforming to PPP-B-621, having approximate inside dimensions of 14-1/4 by 14-1/4 by 7-3/8 inches. The canisters shall be separated by interlocking partitions fabricated of fiberboard conforming to type CF, class weather resistant, variety SW, grade W6c of PPP-F-320. Fiberboard pads fabricated of the same material as the interlocking partitions shall be placed on the top, bottom, sides, ends, and between layers, with additional pads as required to make a tight pack. The wood box shall be closed and strapped in accordance with the appendix of the box specification. Strapping conforming to type I, class B of QQ-S-781, shall be used.

5.2.2 Level B. Eighteen canisters, packaged as specified in 5.1, shall be packed the same as that specified in 5.2 except the shipping container shall conform to style RSC, grade V3c fiberboard of PPP-B-636. The fiberboard box shall be closed in accordance with method V closure requirements and reinforced in accordance with the reinforcing requirements in the appendix to PPP-B-636.

5.2.3 Commercial. Canister, packaged as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Levels A and B. In addition to any special marking required by the contract or order, unit packs shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Marking shall be in accordance with MIL-STD-1188.

5.4 Repair parts. When this item is procured for storage and issue as a repair part, preservation, unit packing, packing and marking shall be as specified on the repair part packaging data sheet which is identified by the National Stock Number for the repair part.

6. NOTES

6.1 Intended use. This specification covers the M11 Canister which is intended for use as a component of the M9A1 Protective Mask.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing.
(c) The laboratory to which samples are to be sent.
(d) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Canister resistance. The resistance of the filled and closed canister sorbent container assembly should be 35 ± 5 mm. of water in order to meet the final canister resistance requirement.

6.4 Canister closing. The assembly line should be so arranged that the charcoal and the unprotected canisters, after filling, are exposed to the atmosphere for the shortest possible length of time. Immediately after filling and closing the canisters should be sealed by suitable closures, (Dwgs. B5-3-532 and C5-3-527), placed in air-tight containers such as charcoal drums, or other adequate containers, to prevent undue moisture absorption, pending placing the canisters in the appropriate packaging container (Dwg. D5-41-34) or as otherwise specified. These precautions are necessary because charcaol can absorb sufficient moisture during manufacturing operations to cause the canisters to be rejected if proper control is not maintained.

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